INCH-POUND

MIL-P-49452(CR)

6 November 1989

MILITARY SPECIFICATION
Power Supply
18 MM, MICROCHANNEL WAFER, GEN III, MX-10130()/UV

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Power Supply for use in the 18 millimeter Microchannel Wafer, Image Intensifier, MX-10130()/UV.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.f).

SPECIFICATIONS

MILITARY

MIL-P-11268	- Parts, Materials, and Processes Used-in Electronic Equipment
MIL-B-49030/6	- Battery, Dry BA-3058/U
MIL-B-49430/4	- Battery, Primary Lithium, Sulfur-Dioxide BA-5567/U

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A: FSC 5855

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MIL-I-49453 - Image Intensifier Assembly, 18mm Microchannel Wafer MX-10130()UV

STANDARDS

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-781 - Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-810 - Environmental Test Methods
MIL-STD-883 - Test Methods and Procedures for Microelectronics

HANDBOOKS

MILITARY

MIL-HDBK-781 - Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

7A	7B	
A3140760	A3144261	- Goggle, Assembly
A3140861	A3144308	- Image Intensifier Assembly, 18mm, MX-10130()/UV
A3140865	A3144347	- Power Supply, 18 mm, Microchannel Wafer for MX-10130()/UV

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The Power Supply for the Image Intensifier Assembly, 18 millimeter Third Generation Microchannel Wafer, MX-10130()/UV, herein after referred to as the power supply, shall be a solid state electronic device to convert an unregulated 2.0 to 3.0 volts direct current (Vdc) input to multiple high voltage dc outputs.

3.2 Construction. Unless otherwise specified all parts, materials, and processes used in construction of the power supply shall be in accordance with A3140865 (AN/PVS-7A), A3144347(AN/PVS-7B).

3.2.1 Weight. The power supply shall not exceed 30 grams in its total weight.

3.3 Qualification. Power supplies furnished under this specification shall be products which are authorized by the Government as qualified products (see 3.12.18).

3.3.1 Initial production testing. When specified in the contract or purchase order, the contractor shall furnish the IPT units in accordance with 4.3 (see 6.2.b).

3.4 Materials, parts and processes. The selection of the class, grade or type shall be such that the part material will perform its intended function when assembled in the power supply. Parts and materials not specified shall be selected by the contractor and shall be subjected to all the provisions of MIL-P-11268 and as specified herein. Hybrid Microelectronics or integrated electronics shall be tested to the Class B level of Method 5008/5004 of MIL-STD-883.

3.5 Components. The power supply shall consist of a multiplier subassembly and an oscillator subassembly.

3.6 Operation characteristics. Unless otherwise specified herein, the power supply shall have the following performance characteristics at ambient conditions (see 3.12.1). The power supply shall meet all operational characteristics when powered by battery BA-5567/U conforming to MIL-B-49430/4 and battery BA-3058/U conforming to MIL-B-49030/6 or by an equivalent 2.0 to 3.0 Vdc power source.

3.6.1 Adjustments.

3.6.1.1 Voltage adjustment range. The microchannel plate input voltage (V_2) shall be variable by adjusting R_1.

a. V_2 shall be less than 650 Vdc when R_1 is varied counterclockwise with the following conditions applied to the power supply):

(1) Input voltage is 3.0 Vdc.

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(2) The load from V_2 to ground (V_4) is 900 megohms.

(3) The photocathode voltage V_1 and the screen voltage V_3 shall be at no load (see 3.12.6).

b. V_2 shall be greater than 950 Vdc when R_1 is varied clockwise with the following conditions applied to the power supply:

(1) Input voltage is 2.0 Vdc.

(2) The load from V_2 to V_4 is 150 megohms.

(3) V_1 and V_3 shall be at no load.

3.6.1.2 ABC adjustment range. R_2 shall adjust the ABC finish current, I_f over the range from 30 to 100 nanoamperes (nA), as a minimum. The requirement for the ABC start current I_s of 3.6.2.2 shall be met for all positions at R_2 within this range.

3.6.2 Automatic brightness control (ABC).

3.6.2.1 ABC finish current, I_f. V_2 shall be limited by adjustment R_2 to less than 100 Vdc for load currents through V3 over the current range from 30 to 100 nanoamperes (nA). I_f shall not vary more than $\pm$ 10 percent when the input voltage is changed from 3.0 to 2.0 Vdc.

3.6.2.2 ABC start current I_s. V_2 shall increase to no less than 90 $\pm$ 2 percent of its preset value for a 2nA to 20 percent or 8nA (whichever is greater) reduction of the load current through V_3 below the preset I_f current.

3.6.3 Adjustment characteristics.

3.6.3.1 Voltage adjustment linearity. V_2 shall increase as R_1 is varied in a clockwise direction. The voltage adjustment shall be continuous and linear to within 25 percent. When R_1 is at its maximum full clockwise setting V_2 shall be not greater than 1300 Vdc.

3.6.3.2 ABC adjustment linearity. I_f shall decrease as R_2 is varied in a clockwise direction. The current adjustment shall be continuous and linear to within 25 percent.

3.6.4 Output regulation.

3.6.4.1 Normal regulation mode. (see 3.12.15)

a. V_1 shall be as specified in Table I.

b. V_2 shall be as specified in Table I.

c. V_3 shall be as specified in Table I.

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3.6.4.2 ABC mode regulation.

a. V_1 shall have a source impedance of 8 to 18 gigohms.

b. V_2 shall be less than 92 percent of its initial voltage setting of 3.6.4.1.b for load currents through V3 greater than I_s.

c. V_3 shall be 5700 +300, -900 Vdc from 30 percent to 125 percent of the ABC finish current.

3.6.4.3 Bright source protection mode regulation.

a. V_1 shall be 32 $\pm$ 4 Vdc at 0.2 megohm load from V_1 to V_2. At 90 megohm load from V_1 to V_2, V_1 shall be 32 +6, -4 Vdc.

b. V_2 shall be less than 100 Vdc for load currents through V_3 greater than or equal to I_f.

c. V_3 shall be 5700 +300, -900 Vdc from 30 percent to 125 percent of the ABC finish current.

3.6.5 Input current characteristics.

3.6.5.1 Maximum peak starting (MPS) and maximum peak to peak steady state (MPPSS). The MPS input current shall not exceed 200 milliamperes (mA) and the MPPSS input current shall not exceed 200 mA.

3.6.5.2 The MASS input current. The maximum average steady state (MASS) input current shall not exceed 22mA.

3.6.6 Rise time. The rise time of the power supply shall be not greater than 3 seconds when the power supply is operated with the intensifier module MX-10130()/UV. The voltage overshoot of V_2 shall not exceed 20 Vdc over the normal mode steady state value.

3.6.7 Response time. When the power supply is operating the intensifier module MX-10130()/UV which is subjected to a step change in input illumination level, its response time shall be less than 1.5 seconds. The response time of the power supply to a transient illumination level shall be less than 2.0 seconds. The voltage overshoot of V_2 shall be not greater than 20 Vdc above its normal mode steady state value.

3.6.8 Ripple voltage. The peak to peak ripple voltage shall not exceed 3 percent of its steady state value for V_1 and V_3 and 50 volts for V_2 when the power supply is operated in the normal mode. The frequency of the ripple voltage shall be greater than 20 kHz.

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3.6.9. Operational stability.

a. The power supply shall not exhibit arcing. There shall be no evidence of fluctuation or oscillation of the MASS input current greater than ± 0.5 mA.

b. V2 shall not fluctuate greater than ± 5 Vdc from its steady state value and shall not drift greater than ± 5 Vdc from its steady state value. Periodic fluctuations greater than the above requirements shall be allowed at the V_2 oscillator frequency but shall not exceed the ripple voltage requirement (see 3.6.8).

3.6.10 Operational safety.

3.6.10.1 Short circuit protection. While operating, the power supply shall not be damaged (see 3.12.4) when subjected to a direct short circuit of the high voltage output terminals to ground for at least one minute.

3.6.10.2 Reverse polarity. The power supply shall withstand a reverse polarity input voltage of 3.0 Vdc for at least 10 minutes. When the correct polarity of 3.0 Vdc input is applied to the power supply, the power supply shall conform to the requirement of 3.6.5.2.

3.6.10.3 Ground probe. While operating, the power supply shall not be damaged by ground probing.

3.6.10.4 Overvoltage protection. The power supply shall operate in the normal regulation mode at 23°C (see 3.6.4.1) for at least 10 minutes when powered by an input voltage of 4.0 Vdc.

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TABLE I. Output regulation, normal mode.

Output Voltage	Temperature Minus (-) 51°C	Temperature Plus (+) 23°C	Temperature Plus (+) 49°C
V_1	600, 700, 800, 900 (+25 -75) Vdc at no load	600, 700, 800, 900 (+25 -75) Vdc at no load	600, 700, 800, 900 (+25 -75) Vdc at no load
V_2	1. Within +30, -40 Vdc of the initial preset value set at +23°C. The V_2 shall be 200 ± 2% of the load at +23°C (V_1 and V_3 at no load). 2. Within +30, -60 Vdc of the initial preset value at +23°C as the value current of V_3 varies from 0 to 30% of the ABC finish current.	1. Within ± 20 Vdc of the initial preset value as the load current value* varies from 0 to 30% of the ABC finish current.	1.Within +30,-40 Vdc of the initial preset value set at +23°C. The load on V_2 shall be 80 ±2% of the load at +23°C. (V_1 and V_3 at no load). 2.Within +30, -60 Vdc of the initial preset voltage as the load current at V_3 varies from 0 to 30% of the ABC finish current.
V_3	1. 5700 ± 300, Vdc for a load current 0.3 I_f.	1. 5700 ± 300, Vdc for a load current 0.3 I_f.	1.5700 ± 300, Vdc for a load current 0.3 I_f.

* NOTE: Acceptable alternate V_1 values shall be as specified in procurement documents. The initial preset value shall be determined by adjusting R_1 and R_2 at the specified load conditions to satisfy the specified power supply operating setting.

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3.6.11 Electromagnetic interference. The power supply assembly properly incorporated within a MX-10130()/UV image intensifier tube (meeting the requirements of MIL-I-49453) properly incorporated with in a AN/PVS-7A (A3140861) AN/PVS-7B (A3144308) shall comply with the requirements of section 3.6.8 of the system specifications.

3.7 Environmental conditions.

3.7.1 Temperature operation. The power supply shall not be damaged, shall operate throughout the range of -51°C to +49°C, and shall meet the following performance requirements at the temperature specified. The power supply shall also meet the operation characteristics when powered by battery BA-5567/U, BA-3058/U or by an equivalent 2.0 to 3.0 Vdc voltage source as shown in Figure 1.

3.7.1.1 Temperature operation at +49°C.

a. Output regulation.

(1) Normal mode. The output regulation requirements shall be as specified in Table I.

(2) ABC mode. The output regulation requirements shall be as specified in 3.6.4.2.

(3) Bright source protection mode. The output regulation requirements shall be as specified in 3.6.4.3.

b. For all discrete positions of the adjustment R_2 meeting the requirement of 3.6.2.1, the ABC finish current at +49°C shall not vary more than ± 30 percent of the corresponding value at ambient condition with 3.0 Vdc input. The ABC start current shall be 2 nA to 20% or 8na (whichever is greater) below the ABC finish current at +49°C.

c. Rise time shall be not greater than 5 seconds. The voltage overshoot of V_2 shall not exceed 20 Vdc of its normal mode steady state value (see 3.6.4.1).

d. The MASS input current shall be not greater than 24 mA.

e. The operational stability requirements of 3.6.9 shall be met.

3.7.1.2 Temperature operation at -17.8°C.

a. The rise time shall be not greater than 5 seconds.

b. Operational stability requirements of 3.6.9 shall be met.

3.7.1.3 Temperature operation at -51°C.

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a. Output regulation:

(1) *Normal mode.* The output regulation requirements shall be as specified in Table I.

(2) ABC mode. The output regulation requirements shall be as specified in 3.6.4.2.

(3) Bright source protection mode. The output regulation requirements shall be as specified in 3.6.4.3.

b. For all discrete positions of the adjustment R_2 meeting the requirement of 3.6.2.1, the ABC finish current at -51° C shall not vary more than ± 30 percent of the corresponding value at the ambient condition with 3.0 Vdc input. The ABC start current shall be 2nA to 20 percent or 8nA (whichever is greater) below the ABC finish current at -51° C.

c. The rise time shall be not greater than 5 seconds. The voltage overshoot of V_2 shall not exceed 20 Vdc of its normal mode steady state value (*see* 3.6.4.1).

d. The MASS input current shall be not greater than 20 mA.

e. Operational stability requirements of 3.6.9 shall be met.

3.7.2 Environmental temperature stress. The power supply shall neither be damaged nor exhibit degradation of performance after the storage or thermal shock environments specified herein.

3.7.2.1 High temperature stress. Storage at +71° C.

3.7.2.2 Low temperature stress. Storage at -51° C.

3.7.2.3 Temperature shock. Temperature changes occurring within 5 minutes over the temperature range of +71° C to +23° C and +23° C to -51° C.

3.7.3 Mechanical vibration. The power supply shall not be damaged and shall meet the operational requirements after being vibrated with simple harmonic motion parallel to and perpendicular to the power supply's longitudinal axis over a frequency range of 5 to 55 Hertz (Hz) at an amplitude of not less than 0.10 inch total excursion for 10 minutes in each direction.

3.7.4 Mechanical shock. The power supply shall not be damaged and shall meet the operational requirements after subjection to shock impacts of 75 g's at the peak amplitude applied in each of three mutually perpendicular directions.

3.7.5 Humidity. The power supply shall not be damaged when exposed to 95 percent relative humidity at +20°C to +65°C for prolonged duration.

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3.8 Burn-in/ESS. Each power supply shall be set to the standard operating setting (see 3.12.9) and shall be subjected to seven (7) burn-in test cycles. Unless otherwise specified, no adjustment shall be allowed during or after the burn-in. The burn-in test cycle shall consist of twenty-four duty cycles at 55 minutes on-time and 5 minutes off-time at a temperature of +49° C with operation in the normal regulation mode, followed by 55 minutes on-time at -51° C in the normal regulation mode.

3.9 Reliability. The power supply shall have a lower test (θ_o) mean time between failure (MTBF) of 9,000 hours under conditions specified herein.

3.10 Identification markings. Each power supply shall be marked for identification in accordance with A3140865 (AN/PVS-7A) A3144347 (AN/PVS-7B). The marking shall include a coded acceptance date. The first two numbers of the code shall be the last two digits of the year. The last two numbers shall be two digits indicating the calendar week of the year (01 through 52). Reading from left to right (or top to bottom), the code number shall designate the year and week of acceptance in that order.

3.11 Workmanship. The power supply shall conform to MIL-STD-454, Requirement 9. The power supply shall be free from burrs, sharp edges, potting voids, defects, or scratches which may affect power supply performance. It shall also be free of grease, oil, dust, solder flux, or any other conductive film.

3.12 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.12.1 Ambient conditions. Ambient conditions are defined as follows:

Temperature	23° C, ± 10° C (73° F, ± 18° F
Relative humidity	50 percent, ± 30 percent
Atmospheric pressure	725, +50mm of mercury -75mm of mercury (28.5, +2.0 inches of mercury -3.0 inches of mercury)

3.12.2 ABC finish current. The load current through V_3 that results in a reduction of the absolute value of V_2 to not greater than 100 Vdc.

3.12.3 ABC start current. The load current through V_3 that results in a change of V_2 from its initial value at no load condition to 90 +2 percent of that value.

3.12.4 Damage. Damage is defined as any breakage, loosening, shift, deformation, failure of any finish, hardware, connection or component, change in input current greater than ± 15 percent, corona, arcing, voltage instability, electrical failure or malfunctioning, or any deterioration of quality or failure to operate in accordance with the requirements of the specification.

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3.12.5 "g". "g" is acceleration or deceleration of 32.17 feet per second per second.

3.12.6 No load. The output currents whose values do not exceed 2 nanoamperes for V_2 and V_3, and 1 nanoampere for V_1.

3.12.7 Rise time. The time required for V_2 to settle within 10 volts of its steady state value after the input voltage has been applied.

3.12.8 Steady state. The condition in which each output voltage remains constant after initial transient or fluctuation conditions have disappeared.

3.12.9 Standard operating setting. The standard operating setting shall be obtained as follows:

a. All parameters are shown in Figure 2.

b. Input voltage shall be 3.0, ± 0.05 Vdc.

c. The load between outputs V_2 and V_4 shall be 150 megohms and all other outputs shall be at no load conditions.

d. V_2 shall be equal to 800 ± 150 Vdc.

e. The ABC finish current shall be equal to 60 ± 5 nA.

3.12.10 Low voltage operating setting. The low voltage operating setting shall be obtained as follows:

a. All parameters are shown in Figure 2.

b. Input voltage shall be 3.0, ± 0.05 Vdc.

c. The load between outputs V_2 and V_4 shall be 900 megohms and all other outputs shall be at no load conditions.

d. V_2 shall be equal to or less than 650 Vdc.

e. The ABC finish current shall be equal to or less than 30 nA.

3.12.11 High voltage operating setting. The high voltage operating setting shall be obtained as follows:

a. All parameters are shown in Figure 2.

b. Input voltage shall be 3.0, +0.05 Vdc.

c. The load between outputs V_2 and V_4 shall be 150 megohms and all other outputs at no load conditions.

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d. V_2 shall be equal to or greater than 950 Vdc.

e. The ABC finish current shall be equal to or greater than 100 nA.

3.12.12 Room temperature. Room temperature is defined as 23°C, ± 3°C.

3.12.13 Photocathode current. A signal induced current at V_1 when the cathode of MX-10130()/UV is illuminated.

3.12.14 Response time. The time for the V_2 voltage to be within 10 volts of its steady state output voltage.

3.12.15 Normal regulation mode. The power supply operating condition in which all voltages are within the tolerances of their specified operating setting.

3.12.16 ABC regulation mode. The power supply operating condition which provides automatic brightness control to the intensifier tube by reducing voltages V_1 and V_2 from their initial specified operating setting as a result of load currents induced through V_1 and V_3.

3.12.17 BSP regulation mode. The power supply operating condition which provides bright source protection to the intensifier tube from excessive current by reducing voltages V_1 and V_2 from their initial specified operating setting to very low voltage levels as a result of load currents induced through V_1 and V_3.

3.12.18 Qualified product. A product which has successfully met all of the requirements of an IPT or first article witnessed by the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract.

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Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Materials inspection. The supplier is responsible for parts and materials used in manufacturing the power supply in accordance with referenced specifications, standards, and drawings as specified herein.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Initial production testing (4.3).
b. Quality conformance inspections (4.5).

4.3 Initial production testing. Unless otherwise specified in the contract or purchase order the contractor shall furnish 15 power supplies for IPT (see 6.2.b).

4.3.1 Inspection. Each IPT power supply shall be examined in accordance with Table II. Presence of one or more defects shall be cause for rejection of that power supply.

TABLE II. Inspection.

Requirements	Requirement Paragraph
Components missing or not as specified	3.5
Construction not as specified	3.2
Identification or special marking missing or illegible	3.10
Workmanship not as specified	3.11, 3.4

4.3.2 Tests.

4.3.2.1 Initial production testing, subgroup 1. Following successful completion of the inspection specified in 4.3.1, each IPT power supply shall be subjected to all tests of Table III, subgroup 1. The tests of Table III, subgroup 1 shall be performed in the order listed. Unless otherwise specified, the tests of Table III, subgroup 1 shall be performed with the power supply at the standard operating setting (see 3.12.9). Failure of any power supply to pass all tests in Table III, subgroup 1 shall be cause for rejection of that power supply and may be cause for termination of IPT.

4.3.2.2 Initial production testing, subgroup 2. Five power supplies which have successfully passed the tests specified in 4.3.2.1 shall be subjected to the tests of Table III, subgroup 2, at the high voltage operating setting (see 3.12.11).

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An additional 5 power supplies which have successfully passed the tests specified in 4.3.2.1 shall be subjected to the tests of Table III, subgroup 2, while at the low voltage operating setting (see 3.12.10). Humidity shall be tested last. Failure of any power supply to pass all tests in Table III, subgroup 2 shall be cause for rejection of that power supply and may be cause for termination of IPT.

4.3.2.3 Initial production testing, subgroup 3. Two power supplies which have successfully passed the test specified in 4.3.2.1 shall be subjected to all tests of Table III, subgroup 3, while at the standard operating setting. Two additional power supplies may be selected for EMI testing. Failure of any power supply to pass all tests in Table III, subgroup 3 shall be cause for rejection of that power supply and may be cause for termination of IPT.

4.3.2.4 Initial production testing, subgroup 4. Ten power supplies which have successfully passed the tests specified in 4.3.2.1 and 4.3.2.2 shall be subjected to the reliability tests of Table III, subgroup 4. Failure to meet the accept/reject criteria of the reliability test shall be cause for failure of the reliability test and may be cause for termination of IPT.

TABLE III. Initial production testing.

Inspection	Requirement Paragraph	Test Paragraph
Subgroup 1		
Burn-in	3.8	4.8
Automatic brightness control*	3.6.2	4.6.2
Output regulation*	3.6.4	4.6.4
Input current characteristics*	3.6.5	4.6.5
Rise time*	3.6.6	4.6.6.1
Response time*	3.6.7	4.6.7
Operational stability*	3.6.9	4.6.9
Operational safety*	3.6.10	4.6.10
Temperature operation	3.7.1	4.7.1
Voltage adjustment range *	3.6.1.1	4.6.1.1
Automatic brightness control adjustment range*	3.6.1.2	4.6.1.2
Adjustment characteristics*	3.6.3	4.6.3

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TABLE III. Initial production testing. -(Continued)

Inspection	Requirement paragraph	Test paragraph
Subgroup 2		
Environmental temperature stress	3.7.2	4.7.2
Mechanical vibration*	3.7.3	4.7.3
Mechanical shock*	3.7.4	4.7.4
Humidity	3.7.5	4.7.5
Subgroup 3		
Temperature operation at -17.8°C	3.7.1.2	4.7.1.2
Ripple voltage*	3.6.8	4.6.8
Electromagnetic interference*	3.6.11	4.6.11
Subgroup 4		
Reliability	3.9	4.9

* Test performed at room temperature (see 3.12.12).

4.4 Test conditions. Test shall be conducted with the test procedures specified herein. Unless otherwise specified the following conditions shall apply throughout the specification.

a. Tests shall be performed at ambient conditions.

b. The applied input voltage shall be 3.0 volts dc.

c. Tolerance on applied input voltages shall be ± 0.05 volts dc.

d. Test chambers used for environmental temperature tests shall maintain the temperature within ± 2° C.

e. All meters used during tests shall have been calibrated within 30 days prior to the start of testing, and thereafter as required, and at intervals not to exceed six months. All meters shall be accurate within 1.0 percent of full scale reading, except the meter used to measure V_2 which shall be accurate within 0.5 percent of full scale reading.

f. Tests shall be performed with the power supply operated in an approved simulated grounded tube test housing (Figure 3). Measurement shall be performed utilizing the power supply test circuit (Figure 2), or its equivalent. Unless otherwise specified all switches shall be in the open position except for switch S_5.

g. Tests shall be performed at the standard operating setting.

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h. Resistor tolerance shall be ± 5 percent.

4.5 Quality conformance inspection.

4.5.1 Group A inspection. Each power supply shall be inspected for the attributes as specified in Tables II and IV. The order of tests following burn-in shall be optional. Failure of a power supply to pass all inspections of table IV shall be cause to reject the power supply.

TABLE IV. Group A inspection.

Inspection	Requirement Paragraph	Test Paragraph
Burn-in	3.8	4.8
Automatic brightness control	3.6.2	4.6.2
Output regulation	3.6.4	4.6.4
MASS input current	3.6.5.2	4.6.5.2

4.5.2 Group B inspection. Power supplies which have passed Group A inspections in accordance with 4.5.1 shall be formed into lots and shall be subjected to Group B inspections of table V. Sampling shall be in accordance with the contract or purchase order. Tests for temperature operation at +49° C and at -51° C shall be performed prior to the test for adjustments. At the completion of the Group B inspection, all power supplies shall be at the standard operating setting.

TABLE V. Group B inspection.

Inspection	Requirement Paragraph	Test Paragraph
Temperature oper at +49° C	3.7.1.1	4.7.1.1
Temperature oper at -51° C	3.7.1.3	4.7.1.3
Adjustments	3.6.1	4.6.1
Dimensions		
a. Width	A B	
b. Inside radius	A3140865 A3144347	
c. Outside radius		
d. Location of terminals and leads		

4.5.3 Group C inspection. Group C inspection shall consist of the tests specified in Table VI and shall be performed on power supplies from lots which have passed Group A or A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

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4.5.3.1 Sampling of Group C inspections. Sampling shall be in accordance with the contract or purchase order (see 6.2.c).

4.5.3.2 Group C failures. Actions required in the event of a Group C failure shall be as specified in the contract or purchase order (see 6.2).

4.5.3.3 Reinspection of conforming Group C samples. Unless otherwise specified, the samples which have been subjected to and passed Group C inspection shall be accepted on the contract provided all damages are repaired. Samples which were subjected to Humidity shall be baked prior to use.

TABLE VI. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1 - High voltage		
Operational safety	3.6.10	4.6.10
Environmental temperature stress	3.7.2	4.7.2
Mechanical vibration	3.7.3	4.7.3
Mechanical shock	3.7.4	4.7.4
Humidity	3.7.5	4.7.5
Subgroup 2 - Low voltage		
Operational safety	3.6.10	4.6.10
Environmental temperature stress	3.7.2	4.7.2
Mechanical vibration	3.7.3	4.7.3
Mechanical shock	3.7.4	4.7.4
Humidity	3.7.5	4.7.5

4.5.4 Group D inspection. Group D inspection shall consist of the tests specified in Table VII and shall be performed on units from lots which have been subjected to and met Group B inspection. Test conditions, methods, and procedures shall be in accordance with paragraph 4.9 and MIL-STD-781.

4.5.4.1 Sampling for Group D inspection. Ten power supplies shall be selected at random from each month's production. The power supplies from two lots, a total of 20 power supplies, will be combined to form the reliability sample.

4.5.4.2 Group D failure. Failure of the power supplies on test to meet the requirements of Table VII shall be considered a failure of the Group D inspection. Actions required in the event of a Group D failure shall be specified in the contract or purchase order.

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TABLE VII. Group D inspection.

Inspection	Requirement Paragraph	Test Paragraph
Reliability	3.9	4.9

4.5.4.3 Reinspection of conforming Group C and D samples. Unless otherwise specified the samples which have been subjected to and passed Group C and D inspection shall be accepted on the contract. Units which were damaged or failed Group D inspection shall not be accepted on the contract.

4.6 Test methods.

4.6.1 Adjustments.

4.6.1.1 Voltage adjustment range.

a. With the conditions applied to the power supply as specified in 3.6.1.1a, vary R_1 to a full counterclockwise direction and measure V_2. The setting of R_2 shall not be changed during the performance of this test.

b. With the conditions applied to the power supply as specified in 3.6.1.1b, vary R_1 to a full clockwise direction and measure V_2. The setting of R_2 shall not be changed during the performance of this test.

Failure to meet the requirements of 3.6.1.1 shall constitute failure of this test.

4.6.1.2 ABC adjustment range.

a. With 3.0 Vdc input, measure and record V_2 as the initial preset value.

b. With S_4 closed adjust the ABC current source to not less than 100nA and record the current as I_f.

c. Adjust R_2 until V_2 reaches 100 Vdc or less.

d. Decrease the ABC source current, with R_2 remaining fixed until V_2 equals 90 $\pm$ 2 percent of its initial preset value for step a. Measure and record the current as I_s.

e. Calculate the difference between I_s and I_f.

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f. Reset the ABC source current to not greater than 30nA and record the current as I_f. Repeat steps c through e.

Failure to meet the requirement of 3.6.1.2 shall constitute failure of this test.

4.6.2 Automatic brightness control.

4.6.2.1 ABC finish current, I_f.

a. With 3.0 Vdc input, measure and record V_2 as the initial preset value.
b. With S_4 closed adjust the ABC current source until V_2 reaches 100 Vdc or less, and record the current as I_f.
c. Change the input voltage to 2.0 Vdc and readjust the ABC current source (if necessary) until V_2 reaches 100 Vdc or less. Calculate the difference between finish current values for the 3.0 Vdc and 2.0 Vdc input.

Failure to meet the requirement of 3.6.2.1 shall constitute failure of this test.

4.6.2.2 ABC start current, I_s.

a. With 3.0 Vdc input decrease the ABC source current until V_2 equals 90 ± 2 percent of the reading of 4.6.2.1.a. Measure and record the current as I_s. Calculate the difference between I_s and I_f of 4.6.2.1.b.
b. Repeat step a. above with 2.0 Vdc input, and calculate the difference between I_s and I_f of 4.6.2.1.c.

Failure to meet the requirement of 3.6.2.2 shall constitute failure of this test.

4.6.3. Adjustment characteristics.

4.6.3.1 Voltage adjustment linearity. Operate the power supply with R_1 positioned for obtaining V2 equal to 800 Vdc. Rotate R_1, clockwise, until V_2 equals 950 Vdc and record the number of degrees rotation of R_1 from the 800 Vdc position as $\theta 1$. Rotate R_1 counterclockwise until V_2 equals 650 Vdc and record the number of degrees rotation of R_1 from the 800 Vdc position as $\theta 2$. Calculate linearity as:

$$\text{Percent Linearity} = \frac{2(\theta 1 - \theta 2)}{(\theta 1 + \theta 2)} \times 100$$

The position of R_2 shall not be changed during this test.
Failure to meet the requirement of 3.6.3.1 shall constitute failure of this test.

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4.6.3.2 ABC current adjustment linearity. Operate the power supply with R_2 positioned for obtaining an ABC finish current of 65nA. Rotate R_2 counter-clockwise while adjusting the V_3 load current until an ABC finish current of 100nA is obtained and record the number of degrees rotation of R_2 from the ABC start current reference position as $\theta 1$. Rotate R_2 clockwise while adjusting the V_3 load current until an ABC finish current of 30nA is obtained and record the number of degrees rotation of R_2 from the reference position as $\theta 2$. Calculate linearity as in 4.6.3.1. The position of R_1 shall not be changed during this test.

Failure to meet the requirement of 3.6.3.2 shall constitute failure of this test.

4.6.4 Output regulation.

4.6.4.1 Normal mode. Perform the measurements required for each of the output voltages as specified in Table VIII. Failure to meet the requirements of 3.6.4.1 shall constitute failure of this test.

4.6.4.2 ABC mode. Perform the measurement required for each of the output voltages as specified in Table IX. Failure to meet the requirements of 3.6.4.2 shall constitute failure of this test.

4.6.4.3 Bright source protection mode. Perform the measurements required for each of the output voltages as specified in Table X. Failure to meet the requirements of 3.6.4.3 shall constitute failure of this test.

4.6.5 Input current characteristics.

4.6.5.1 Maximum peak starting (MPS) and maximum peak to peak steady state (MPPSS). Perform the measurements for MPS and MPPSS as specified in Table XI as follows: All power supply outputs shall be completely discharged to ground before performing the test for MPS. Connect the current probe or equivalent to the dc input lead as in Figure 2. Apply the input voltage for the specified test condition and allow the storage scope or equivalent to display the peak current waveform. Sweep ground on the storage scope and read the maximum sweep point above ground as MPS. The MPS displayed within the initial 100 milliseconds after the applied input voltage shall be excluded from the measurements. Read the peak to peak steady state signal as MPPSS. Failure to meet the requirements of 3.6.5.1 shall constitute failure of this test.

4.6.5.2 MASS input current. Maximum average steady state (MASS). Perform the measurements for MASS as specified in Table XI. Failure to meet the requirements of 3.6.5.2 shall constitute failure of this test.

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4.6.6 Rise time.

4.6.6.1 Rise time, initial production testing. Connect the power supply to the intensifier module as shown in Figure 1. The rise time shall be verified by displaying V_2 on a storage oscilloscope or equivalent. Sensitivity of the equipment shall detect a 20 Vdc overshoot. Perform the measurements of V_2 as specified in Table XII. The tests shall be conducted as follows: Apply the illumination to the intensifier. Then apply the specified input potential to the power supply while monitoring V_2. Sweep the steady state voltage trace for V_2. Measure the time for V_2 to settle within 10 Vdc of its steady state value. All outputs from the power supply shall be fully discharged from any previous operation prior to performing the rise time testing. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.6.6.2 Rise time, quality conformance inspection. The rise time shall be verified by displaying V_2 on a storage oscilloscope or equivalent. Sensitivity of the equipment shall detect a 20 Vdc overshoot. Perform the measurements of V_2 as specified in Table XII. The tests shall be conducted as follows: With S_1, S_2, S_3, and S_4 open and R_4 equal to the value required for the specified operating setting, apply the specified input potential to the power supply while monitoring V_2. Sweep the steady state voltage trace for V_2. Measure the time for V_2 to settle within 10 Vdc of its steady state value. All outputs from the power supply shall be fully discharged from any previous operation prior to performing the rise time test. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Response time. Connect the power supply as shown in Figure 1. The response time shall be verified by displaying V_2 on a storage oscilloscope or equivalent. Sensitivity of the equipment shall detect a 20 Vdc overshoot. The rise and fall time of the input voltage to the power supply shall be less than 10% of the pulse width. Apply the input voltage to the power supply and test the response time as follows.

a. While monitoring V_2, apply a step pulsed input illumination to the intensifier module sufficient to cause V_2 to decrease at least 400 Vdc from its initial steady state normal mode voltage. Measure the time for V_2 to settle to its new steady state level. Remove the input illumination and measure the time for V_2 to settle to its steady state normal mode voltage.

b. While monitoring V_2, apply a 1/125 of a second pulse width illumination of the magnitude of a. Measure the combined time for V_2 to fall and return to its steady state normal mode voltage.

Failure to meet the requirements of 3.6.1.2 shall constitute failure of this test.

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TABLE VIII. Output regulation, normal mode tests.

Temperature	Measurements	Switches Closed	Load Current at V_3	3.0 Vdc Input	2.0 Vdc Input
+49° C	V_1, V_2, V_3	——	No load	(*)	*
	V_2	S_4	25 to 30% of ABC finish	(*)	*1, 2
+23° C	V_1, V_2, V_3	——	No load	*	(*)
	V_2	S_4	25 to 30% of ABC finish	*1, 2	(*)
-51° C	V_1, V_2, V_3	——	No load	(*)	*
	V_2	S_4	25 to 30% of ABC finish	(*)	*1, 2

* Test as required

*1,2 Test required for IPT and Group C subgroup 1 and 2 tests only

(*) Test required for IPT only

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TABLE IX. Output regulation, ABC mode tests.

Temperature	Measurements	Switches Closed	Load Current at V_3	3.0 Vdc Input	2.0 Vdc Input
+49° C	V_1, V_2, V_3	S_3, S_4	$.30I_f$	——	(*)
+23° C	V_1, V_2, V_3	S_3, S_4	$.30I_f$	(*)	——
-51° C	V_1, V_2, V_3	S_3, S_4	$.30I_f$	——	(*)

(*) Test required for IPT inspection only.

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TABLE X. Output regulation,bright source protection mode tests.

Temperature	Measurements	Switches Closed	Load Current at V_3	3.0 Vdc Input	2.0 Vdc Input
+49°C	V_1 V_2 V_3	S_1, S_4	I_f	(*) (*) (*)	* (*) (*)
	V_1 V_2	S_2, S_4	I_f I_f	(*) (*)	* (*)
+23°C	V_1 V_2 V_3	S_1, S_4	I_f	* (*) (*)	(*) (*) (*)
	V_1 V_2 V_3	S_2, S_4	I_f	* (*) (*)	(*) (*) (*)
-51°C	V_1 V_2 V_3	S_1, S_4	I_f	(*) (*) (*)	* (*) (*)
	V_1	S_2, S_4	I_f	(*)	(*)

* Test as required

(*) Test required for IPT inspection only

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TABLE XI. Input current characteristics tests.

Temperature	Measurements	Switches Closed	Load Current at V_3	3.0 Vdc Input	2.0VdcInput
+49°C	MASS	——	No load	(*)	*
	MASS	S_1, S_4	I_f to I.25 I_f	(*)	(*)
+23°C	MPS			(*)	——
	MPPSS	——	No load	(*)	——
	MASS			*	(*)
-51°C	MASS	——	No load	(*)	*
	MASS	S_1, S_4	I_f to I.25 I_f	(*)	(*)

* Test as required
(*) Test required for IPT inspection only

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TABLE XII. Rise time test.

Temperature	Illumination Level Equivalent Photocathode current	Voltage Output	3.0 Vdc Input	2.0 Vdc Input
+49°C	0.01 nA(1)	V_2	(*)	*
	0.5 - 1.0 uA	V_2	(*)	(*)
+23°C	0.01 nA	V_2	(*)	(*)
	0.5 - 1.0 uA	V_2	(*)	(*)
-17.8°C	0.01 nA	V_2	(**)	(**)
	0.5 - 1.0 uA	V_2	(**)	(**)
-51°C	0.01 nA(1)	V_2	(*)	*
	0.5 - 1.0 uA	V_2	(*)	(*)

1) Photocathode currents applicable to IPT inspection only; quality conformance inspection shall be conducted without the intensifier module.

* Test as required

(*) IPT test only.

(**) IPT inspection only; input voltage to be provided by battery BA-3058/U or BA-5567/U.

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TABLE XIII. Operational stability test.

Temperature	Measurements	Illumination Level Equivalent Photocathode Current	3.0 Vdc Input	2.0 Vdc Input
+49°C	V_2, MASS Input Current	75 - 150 pA	(*)	(*)
	V_2, MASS Input Current	75 - 100 ua	(*)	*
+23°C	V_2, MASS Input Current	75 - 100 ua	(*)	*
-17.8°C	V_2, MASS Input Current	75 - 100 ua	(**)	(**)
-51°C	V_2, MASS Input Current	75 - 150 pA	(*)	(*)
	V_2, MASS Input Current	75 - 100 ua	(*)	(*)

* Test as required.

(**) IPT inspection only; input voltage to be provided by battery BA-3058/U or BA-5567/U.

(*) Test required for IPT inspection only.

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4.6.8 Ripple voltage. Disconnect all voltmeters from the power supply. With the R_4 at 120 megohms, apply an input voltage of 3.0 Vdc. Display a ripple voltage at each output using a Jennings Probe and storage oscilloscope or equivalent. Read the peak to peak trace as the ripple voltage. Calculate percent ripple voltage as follows:

$$\frac{V_{Ripple\ p\text{-}p}}{\text{Vdc output}} \times 100\%$$

Measure the period (T) of the waveform of the displayed ripple voltage for V_2 and V_3 and calculate the frequencies as 1/T. Failure to meet the requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Operational stability. Connect the power supply to the intensifier module as shown in Figure 1. The operational stability shall be verified by (a) observing for arcing and monitoring MASS input current, and (b) displaying V_2 on a storage oscilloscope or equivalent for a period of not less than 60 seconds. Sensitivity of the equipment shall detect 0.5mA and 5 Vdc fluctuations. Perform the measurements specified in Table XIII for input current and V_2 as follows. Apply the specified input voltage to the power supply. Apply the specified step change of input illumination to the photocathode of the intensifier module, and within 2 seconds monitor V_2 and MASS current for not less than 60 seconds. The rise time of the input illumination step changes shall be less than 250 milliseconds. If the voltage fluctuation or drift is greater than plus or minus 5 volts but less than plus or minus 10 volts from a steady state condition during the one minute monitoring period, the test shall be continued an additional 2 minutes during which the requirement of paragraph 3.6.9(b) shall be met. Failure to meet the requirements of 3.6.9 shall constitute failure of this test.

4.6.10 Operational safety.

4.6.10.1 Short circuit protection. Operate the power supply. While operating, directly short all outputs to ground for at least one minute. Remove the short and inspect the power supply for damage. Failure to meet the requirements of 3.6.10.1 shall constitute failure of this test.

4.6.10.2 Reverse polarity. Apply a reverse polarity of 3.0 volts dc input to the power supply for at least 10 minutes. Failure to meet the requirements of 3.6.10.2 shall constitute failure of this test.

4.6.10.3 Ground probe resistance. Mount the power supply optimally. With the power supply operating, scan the surfaces of the power supply, with a 1/8 inch diameter ground probe. During the probing of the power supply, inspect for damage by monitoring the input current and output voltages. Failure to meet the requirements of 3.6.10.3 shall constitute failure of this test.

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4.6.10.4 Overvoltage protection. Operate the power supply with an input voltage of 4.0 Vdc for at least 10 minutes in the normal regulation mode at 23°C. While operating, measure V_1, V_2, and V_3. Failure to meet the requirements of 3.6.10.4 shall constitute failure of this test.

4.6.11 Electromagnetic interference. The AN/PVS-7A (A3140760), AN/PVS-7B (A3144261) under test shall consist of a binocular assembly powered by a battery. In performing the RS03 test, the system shall be oriented in two (2) different positions. These positions will be designated as follows:

a. Horizontal 0° - Objective lenses facing source of radiation
b. Horizontal 270° - Rotated clockwise 270° from horizontal 0°

4.6.11.1 EMI tests. The tests to be performed will be RE02, RS02 and RS03 and will be performed per paragraph 3.6.8 of the systems specifications.

4.6.11.2 RS03 testing. During the RS03 testing, the output brightness shall be monitored continuously and recorded on each monocular assembly. The system shall be operated in a linear operating region (5×10^{-5} FC photocathode illumination), in an ABC region (1×10^{-4} FC to 1×10^{-3} FC photocathode illumination). In performing the RS03 testing the illumination on the photocathode does not have to be a color temperature of 2856°K. However, the illumination levels shall provide the operating modes of linear and ABC as referred above. The U.U.T shall be scanned with both the horizontal and vertical polarized fields.

4.7 Environmental conditions.

4.7.1 Temperature operation.

4.7.1.1 Temperature operation at +49° C. Place the power supply in a test chamber at room temperature (see 3.12.12). Raise the temperature of the test chamber to +49° C and hold at this temperature for a minimum of one hour. At the end of the one hour stabilizing period, the following tests shall be performed.

a. ABC current. The test procedure for ABC current shall be in accordance with 4.6.2.

b. Output regulation as specified in Tables VIII, IX, and X. The test procedure for output regulation shall be in accordance with 4.6.4.

c. MASS input current as specified in Table XI. The test procedures for input current shall be in accordance with 4.6.5.2.

d. Rise time as specified in Table XII. The test procedure for rise time shall be in accordance with 4.6.6.

e. Operational stability as specified in Table XIII. The test procedure for operational stability shall be in accordance with 4.6.9.

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Failure to meet the requirements of 3.7.1.1 shall constitute failure of this test.

4.7.1.2 Temperature operation at -17.8° C. After the +49° C tests are completed, remove the input voltage to the power supply. Lower the temperature of the test chamber to room temperature and place a BA-5567/U or BA-3058/U in the test chamber. Lower the temperature of the test chamber to -17.8° C. Hold this temperature for a minimum of one hour. At the end of this one hour period, measure the rise time and operational stability specified in Tables XII and XIII when the BA-5567/U or BA-3058/U is used as the input voltage source to the power supply. Failure to meet the requirements of 3.7.1.2 shall constitute failure of this test.

4.7.1.3 Temperature operation at -51° C. After testing at +49° C and -17.8° C, remove the input voltage to the power supply and lower the temperature of test chamber to -51° C, and hold at this temperature for a minimum of one hour. At the end of the one hour stabilizing period, perform the tests for output regulation, ABC current characteristics, rise time, MASS input current, and operational stability as specified in Tables VIII through XIII using the same test procedures as 4.7.1.1. At the conclusion of the tests, remove the input voltage to the power supply. Failure to meet the requirements of 3.7.1.3 shall constitute failure of this test.

4.7.2 Environmental temperature stress.

4.7.2.1 High temperature stress. Test the power supply in accordance with MIL-STD-810, Method 501.1. The following details shall apply:

a. Procedure number: Procedure I

b. Pretest performance record: Output regulation (4.6.4), ABC current characteristics (4.6.2), MASS input current (4.6.5.2).

c. Performance record at temperature - Step 5: Temperature operation at +49° C (4.7.1.1).

d. Step 2: Storage temperature at +71° C.

e. Step 3: Storage time - Not less than 2 hours.

f. Step 4: Operating temperature at +49° C.

g. Steps 7 and 8 to be omitted.

Failure to meet the requirements of 3.7.2.1 shall constitute failure of this test.

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4.7.2.2 Low temperature stress. Test the power supply in accordance with MIL-STD-810, Method 502.1. The following details and exceptions shall apply:

a. Pretest performance record: Output regulation (4.6.4), ABC current characteristics (4.6.2), MASS input current (4.6.5.2).

b. Performance record at temperature - Step 5: Temperature operation at -51° C (4.7.1.3)

c. Storage temperature - Step 2: -51° C for not less than 2 hours.

d. Step 3: Not applicable.

e. Lowest operating temperature - Step 4: -51° C.

f. Steps 7 and 8 to be omitted.

Failure to meet the requirements of 3.7.2.2 shall constitute failure of this test.

4.7.2.3 Temperature shock. Test the power supply in accordance with MIL-STD-810, Method 503.1. The following details and exceptions shall apply:

a. Pretest performance record: Output regulation (4.6.4), ABC current characteristics (4.6.2), MASS input current (4.6.5.2), operational stability (4.6.9).

b. Temperature extremes: +71° C to +23° C; +23° C to -51° C.

Failure to meet the requirements of 3.7.2.3 shall constitute failure of this test.

4.7.3 Mechanical vibration. Prior to the start of vibration tests, a pre-test performance record shall be obtained for the power supply of output regulation (4.6.4), ABC current characteristics (4.6.2) and MASS input current (4.6.5.2). Mount the power supply to a test fixture and subject to harmonic motion applied in each of three orthogonal planes; the frequency of the harmonic motion shall be varied logarithmically from 5 to 55 Hz to 5 Hz in one minute; the amplitude shall be 0.05 inch ± .005 inch double amplitude; this frequency shall be repeated 10 times in each plane. The power supply shall not be operated during vibrations. After vibration, a performance test shall be obtained and comparison made with the pretest performance record. Failure to meet the requirements of 3.7.3 shall constitute failure of this test.

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4.7.4 Mechanical shock. Test the power supply in accordance with MIL-STD-810, Method 516. The following details and exceptions shall apply:

a. Procedure number: Procedure 1.

b. Pretest performance record: Output regulation (4.6.4), ABC current characteristics (4.6.2), MASS input current (4.6.5.2), and operational stability (4.6.9).

c. Shock pulse selection: 75 g's peak amplitude, 11 milliseconds nominal duration in accordance with Figure 516-1 of Method 516.

d. Temperature extremes: Not applicable.

e. Operation during tests: Non-operating.

Failure to meet the requirements of 3.7.4 shall constitute failure of this test.

4.7.5 Humidity test method. Subject the unprotected power supplies to the temperature/humidity profile of MIL-STD-810, Method 507.1, Procedure II. After not more than four hours at room temperature, remove the test item from the chamber and perform the following measurements: ABC current characteristics (4.6.2), output regulation (4.6.4), MASS input current (4.6.5.2), rise time (4.6.6), and operational stability (4.6.9). Prior to measurements, excess moisture may be removed from the exterior surface by wiping the external surface only. Failure to meet the requirements of 3.7.5 or the above tests shall constitute failure of this test.

4.8 Burn-in/ESS. Subject the power supply to 7 (seven) burn-in test cycles as specified in 3.8.1. Operation shall be in the normal regulation mode with V_1 and V_3 at no load, and V_2 with R_4 equal to 120 megohms. The transition time from +49° C to -51° C and from -51° C to +49° C shall take place in no less than 15 minutes. The applied input voltage of 3.0 Vdc shall be removed while the temperature is returned to +49° C, and for 1 hour after reaching +49° C, to allow condensation to be baked out of the temperature chamber. At the conclusion of the third cycle, the power supply shall be returned to +49° C for a final 1-hour bake-out. Verification of this test shall be accomplished by audit of the burn-in record.

4.9 Reliability.

4.9.1 Reliability testing. The initial production and quality conformance reliability inspections shall be conducted in accordance with MIL-STD-781 and Test Plan VI-D of MIL-HDBK-781.

4.9.2 Reliability test procedure. The power supply shall be set at the high voltage operating setting prior to subjection to a reliability test cycle as specified in 4.9.2.1. Sufficient reliability test cycles shall be performed to satisfy the requirements of the applicable test plan.

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The number of operating hours for all units shall be continuously monitored and recorded. Failure of a power supply to meet the requirements of Table XIV shall be considered a failure and the hours accumulated by the power supply to the last successful check shall be used in the total hours for the lot under test. At the conclusion of the test, each power supply shall be submitted to and pass the inspection of Table XIV. In addition, during IPT reliability testing, each power supply shall be submitted to the inspection of Table XIV at least once every 400 hours.

4.9.2.1 Reliability test cycle. A reliability test cycle shall consist of 24 duty cycles of 55 minutes on-time and 5 minutes of off-time at no less than +49° C followed by 24 repetitive stress cycles of high light equivalent stress (4.9.2.1.1) and 24 repetitive stress cycles of bright source equivalent stress (4.9.2.1.2) at +49° C, followed by 55 minutes on-time at -51° C in the normal regulation mode. The input voltage shall be 3.0 Vdc and the load from V_2 to V_4 shall be 120 megohms during normal regulation mode operation. The transition time from +49° C to -51° C and from -51° C to +49° C shall take place in no less than 15 minutes. The applied input voltage of 3.0 Vdc shall be removed while the temperature is returned to +49° C, and for 1 hour after reaching +49° C, to allow condensation to be baked out of the temperature chamber. Prior to removing the power supply from test, after exposure to the -51° C temperature, the power supply shall receive a 1-hour bake-out at +49° C.

4.9.2.1.1 High light stress. A cycle of this stress shall consist of the following load conditions sequence.

a. 5 seconds with no greater than 30 gigohm resistor from V_3 to V_4, V_1 at no load, and V_2 at 120 megohms.

b. 10 seconds with a 120 megohm resistor from V_2 to V_4, V_1 and V_3 at no load.

4.9.2.1.2 Bright source stress. A cycle of this stress shall consist of the following load condition sequence.

a. 3 seconds with a 0.2 megohm resistor from V_1 to V_2, V_2 at 120 megohms, and V_3 at no greater than 30 gigohms.

b. 10 seconds with V_2 at 120 megohms, V_1 and V_3 at no load.

TABLE XIV. Reliability performance tests.

Inspection	Requirement Paragraph	Test Paragraph
ABC current characteristics	3.6.2	4.6.2
Output regulation	3.6.4	4.6.4
Operational stability	3.6.9	4.6.9
Mass input current	3.6.5.2	4.6.5.2

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5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The Power Supply is intended to be used in the Image Intensifier, Night Vision, MX-10130()/UV.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Quantity and schedule for IPT testing and disposition of IPT samples for all line items (see 4.3).
c. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling should be conducted per the requirements of MIL-STD-105.
d. Necessary actions by the contractor in the event of a Group C failure (see 4.5.3.2) and/or a Group D failure (see 4.5.4.2).
e. Qualification - If product is not qualified at time of award the contract must require qualification prior to first delivery.
f. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
g. MIL-STD-810C should be used for environmental tests of section 4.

6.3 Definitions. See 3.12.

6.4 Subject term keyword listing.
Power Supply
Image Intensifier Assembly
Microchannel Wafer

Custodian
ARMY-CR

Preparing Activity
ARMY-CR

Project Number 5855-A317

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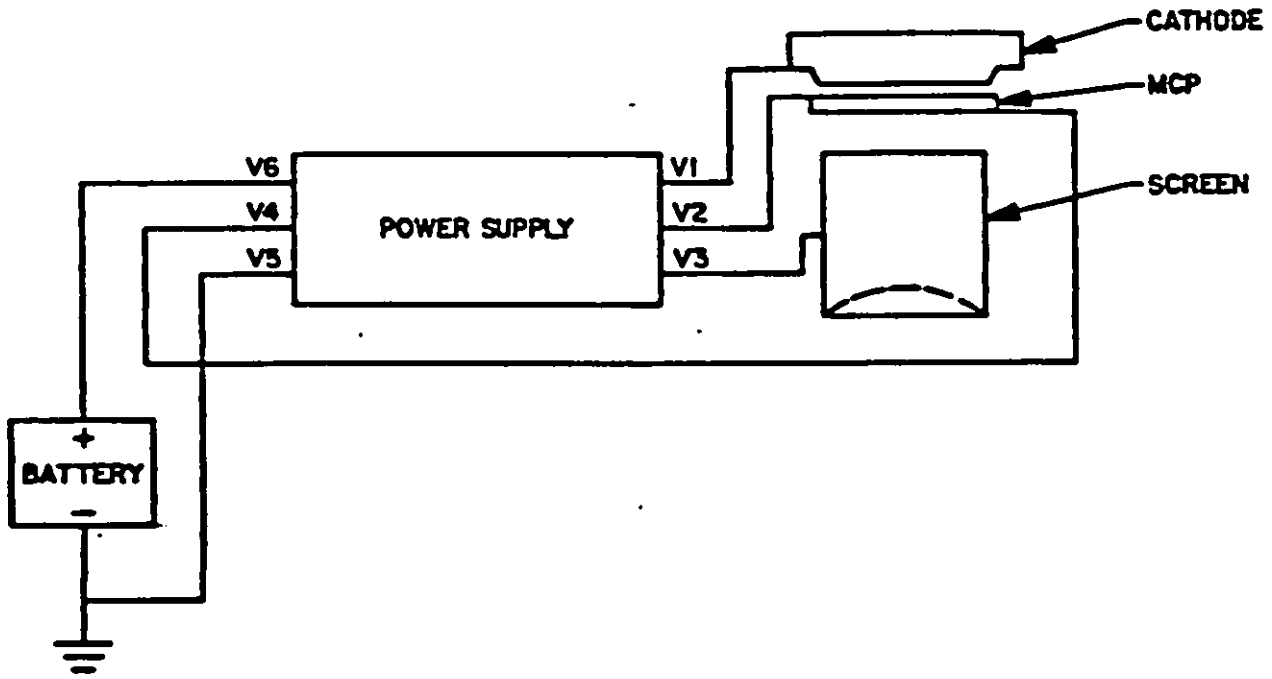


FIGURE 1. WIRING schematic

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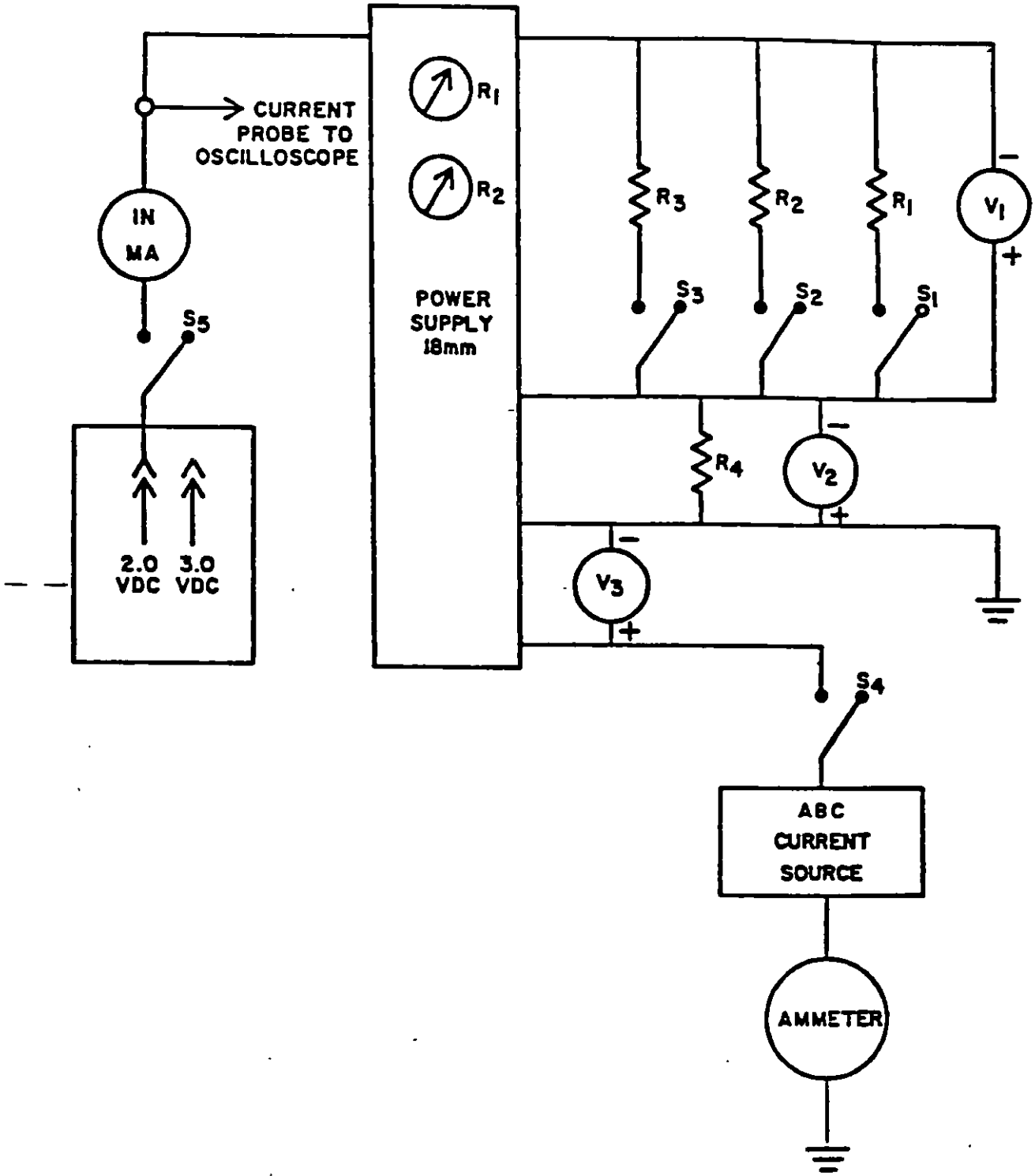
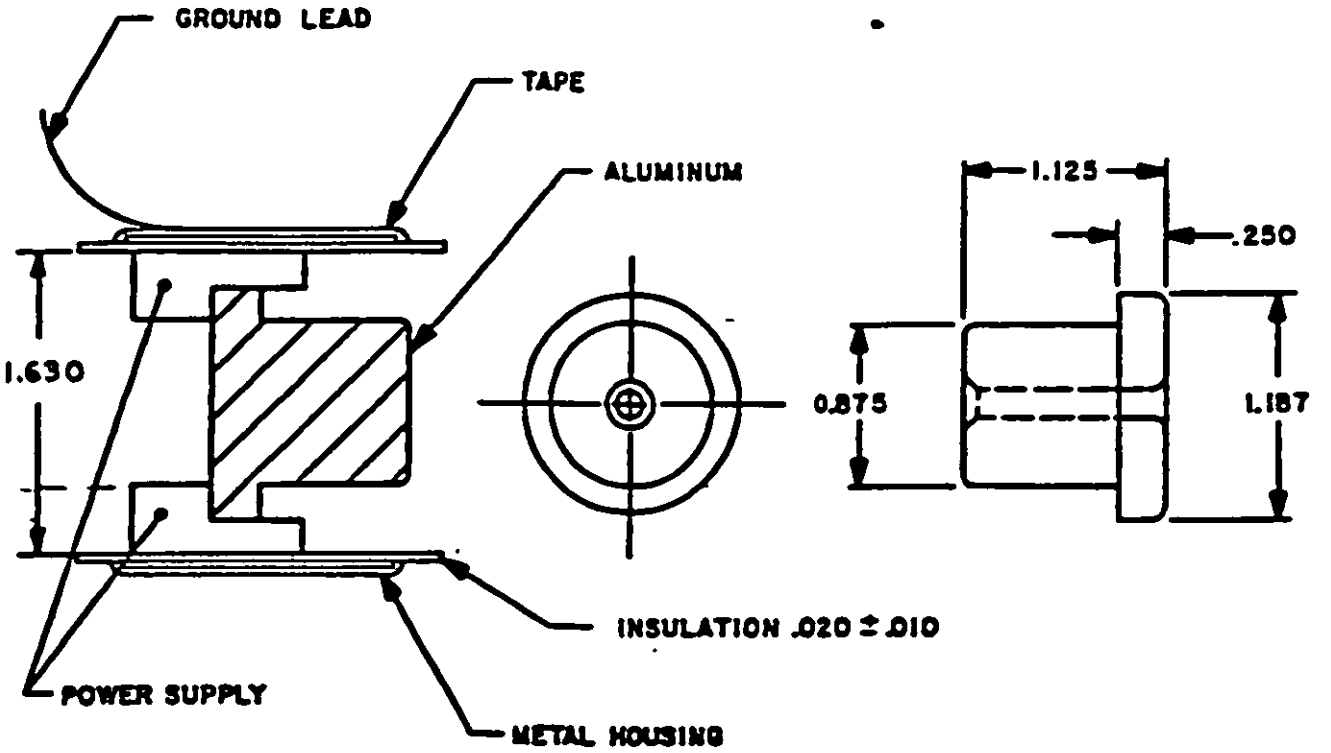


FIGURE 2. Power Supply test circuit

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FIGURE 3. Simulated tube body and housing

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-49452(CR)	Power Supply, 18MM, Microchannel Wafer, Gen III, MX-10130( )/U

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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